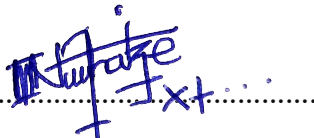


# EE5206: Software Project

## Proposal: Automated Classroom Booking System

### Client Details

|             |                                                                |
|-------------|----------------------------------------------------------------|
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Signature

20.11.2025  
.....

Date

### Group Details

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## 1. Introduction & Problem Statement

University faculties grapple with the complex and continuous task of managing classroom reservations. The current manual or semi-automated systems are often inefficient, resulting in a cascade of operational problems. These include:

- **Frequent Scheduling Conflicts:** Double bookings or booking an occupied room due to a lack of a central, real-time source of truth.
- **Administrative Overhead:** Staff spend a significant amount of time manually processing requests, checking timetables, and resolving conflicts.
- **Poor Resource Utilization:** Rooms may sit empty due to "ghost" bookings, while other lecturers struggle to find space.
- **Lack of Transparency:** Lecturers have no easy way to check room availability, features, or the status of their requests without contacting an administrator.

## 2. Proposed Solution & Objectives

We propose the development of a centralized, intelligent "Automated Classroom Booking System" to solve these problems. The solution will consist of a **robust web application** for administrators and a **user-friendly mobile application** for lecturers and staff.

The primary objectives are to:

- **Automate & Streamline:** Replace manual processes with automated booking/approval workflows and instant conflict detection.
- **Empower Users:** Provide lecturers with a mobile app to search room availability in real-time, submit requests, and receive notifications.
- **Optimize Resources:** Implement priority rules (e.g., exams, large cohorts) and generate utilization reports to help administration make data-driven decisions.
- **Ensure Accountability:** Maintain a complete audit log of all bookings and changes, with role-based access for different user types (Admin, Department, Lecturer).

## 3. Project Scope

The scope of this project is to deliver a complete, self-contained booking platform.

In Scope:

- **Web Application:** An administrative portal to:  
  
Manage room inventory, including adding/editing rooms and their capacities and equipment. Import class timetables to block recurrent slots and define priority rules and blackout periods such as holidays and exam weeks. View, approve, or reject booking requests, and generate analytics and utilization reports.
- **Mobile Application (iOS & Android):** A lecturer/staff-facing app to:

Available rooms can be searched according to date, time, capacity, and equipment requirements. New booking requests can be submitted, and users will receive real-time push notifications such as "Your booking is approved."

#### 4. Features (Functional Requirements)

| Module                             | Key Functional Requirements                                                                                                                                                                                                                                                                                                  |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Admin &amp; Room Mgt.</b>    | <ul style="list-style-type: none"><li>• Admins can set room capacities, equipment tags (projector, A/C), and blackout periods.</li><li>• System shall support "semester rollover" to archive old data and set up a new term.</li></ul>                                                                                       |
| <b>2. Booking &amp; Scheduling</b> | <ul style="list-style-type: none"><li>• Lecturers can search for rooms by date, time, capacity, and equipment.</li><li>• The system shall <b>automatically detect and prevent</b> booking conflicts.</li><li>• The system shall enforce <b>priority rules</b> (e.g., an exam request overrides a regular meeting).</li></ul> |
| <b>3. Workflow &amp; Approvals</b> | <ul style="list-style-type: none"><li>• Booking requests shall trigger an manual approval workflow.</li><li>• Admins can approve or reject requests from the web dashboard.</li><li>• The system shall support multi-level role-based access (e.g., Department Admin, Central Admin).</li></ul>                              |
| <b>4. Notifications</b>            | <ul style="list-style-type: none"><li>• Lecturers shall receive real-time push notifications for booking approval, rejection, or changes.</li><li>• The system shall send email notifications as a secondary method.</li></ul>                                                                                               |
| <b>5. Timetable &amp; Auditing</b> | <ul style="list-style-type: none"><li>• Admins can bulk-import existing faculty timetables (e.g., from a CSV file).</li><li>• The system shall maintain a detailed audit log of all actions (who booked what, when).</li></ul>                                                                                               |
| <b>6. Reporting</b>                | <ul style="list-style-type: none"><li>• Admins can generate utilization reports (e.g., most-used rooms, peak booking hours).</li></ul>                                                                                                                                                                                       |

#### 5. Non-Functional Requirements

- **Performance:** The mobile app and web dashboard must load in short time.
- **Security:** All user data must be encrypted in transit (SSL). The system must be protected against common web vulnerabilities (e.g., SQL Injection, XSS) and enforce strict role-based access control.
- **Usability:** The mobile application must be very user friendly.
- **Reliability:** The system shall have an uptime of 99.9%. All booking data must be backed up regularly to prevent data loss.